Infinite Fourier Tails and High-Mode Coercivity for the Semilocal Weil Quadratic Form

A complete proof of a coupled tail bound for the rank-eight ground state on the support cell $17 \le C \le 19$

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Research preprint. The Riemann Hypothesis is not claimed.

Abstract

Connes, Consani and Moscovici encode the semilocal Weil quadratic form on $[C^{-1/2}, C^{1/2}]$ by an infinite real symmetric matrix $A(C)$ in a Fourier basis, with off-diagonal entries of displacement form $A_{ij} = (b_i - b_j)/(i-j)$. Their spectral programme compares the ground states of finite truncations of $A(C)$ with the zeros of the Riemann zeta function. Any such comparison has to control how the infinite matrix acts on a truncated ground state. We derive the matrix from the explicit-formula distributions and prove three rank-independent estimates. First, an exact identity for the omitted components of $A\,c$, for any finitely supported even vector c, expressed through signed moments of c plus an explicit remainder that is uniform over infinitely many indices. Second, the operator bound $\|A_{\mathrm{off}}\| \le 2\pi \sup_i |b_i|$, obtained from the discrete Hilbert transform. Third, explicit logarithmic coercivity of high-mode compressions. We combine these with validated interval enclosures of the rank-eight even ground vector $u(C)$. The result is the following uniform statement over the whole cell $17 \le C \le 19$: the even components of $A(C)\,u(C)$ at every Fourier index $j > 64$ have joint ℓ^2 norm below 9.491×10^{-12}. Ignoring the signs gives a bound of about 2.906. Moreover, the even compression of $A(C)$ to indices $j \ge 18{,}612{,}929$, and the full compression to $|j| \ge 60{,}879{,}750$, both exceed the identity. Every infinite-index statement is proved analytically; finite-dimensional quantities are certified by outward-rounded ball arithmetic and replayed by an independent implementation. These results do not bound the response of the intermediate modes $9 \le j < 18{,}612{,}929$. They give no uniform profile budget over all ranks and supports, and they do not identify any limit with Riemann's Ξ function. We do not prove the Riemann Hypothesis, and we explain why the present estimates cannot by themselves imply it.

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1 Introduction

1.1 Context

Weil's explicit formula expresses a sum over the nontrivial zeros of ζ as a distribution Ψ built from the pole of ζ, the archimedean place and the primes. Weil's criterion [9] states that the Riemann Hypothesis holds if and only if the quadratic form $f \mapsto \Psi(f^* * f)$ is nonnegative on a suitable class of test functions; see also Bombieri [2]. Restricting the test functions to $[\lambda^{-1}, \lambda] \subset \mathbb{R}_+^*$ produces the *semilocal* form QW_λ. Only the finitely many prime powers $p^a \le \lambda^2$ enter it.

Connes, Consani and Moscovici [3] computed the matrix of QW_λ in the Fourier basis of $L^2([\lambda^{-1}, \lambda], d^*u)$. They observed that the off-diagonal part has displacement form, and constructed from the even ground state of each finite truncation a self-adjoint operator. The spectrum of that operator consists of the real zeros of an entire function. Numerically these zeros approximate the lowest zeros of $\zeta(\frac{1}{2} + is)$ with striking accuracy. Proving convergence would prove RH [3, §7–8].

A necessary ingredient of any convergence argument is control of the *infinite* matrix acting on a *truncated* ground state. If c is the ground vector at retained Fourier rank N, then Ac has infinitely many nonzero components outside the retained block. The retained eigenproblem cannot see these components. A naive row bound on them fails: individual off-diagonal rows decay only like $1/|i-j|$ and are not absolutely summable. This paper proves an estimate of this kind that is uniform over all omitted indices and over a whole support interval.

1.2 Main results

Throughout, $C > 1$ is the support parameter ($C = \lambda^2$ in the notation of [3]), $L = \log C$, and $A = A(C)$ is the infinite Weil matrix of Definition 2.3. The even orthonormal coordinates of a vector are $u = (u_0, \dots, u_N)$, with full coefficients $c_0 = u_0$ and $c_{\pm i} = u_i/\sqrt{2}$. For $n > N$ the even component of Ac at index n is $R_n = \sqrt{2}\,(Ac)_n$.

Main Theorem. Let $17 \le C \le 19$, and let $u(C)$ be the unit ground eigenvector, oriented by $u_0 > 0$, of the even block of the 17×17 truncation of $A(C)$ (retained rank $N = 8$). Then:

(i) **Infinite forcing tail.** $\Big(\sum_{n>64} R_n(C)^2\Big)^{1/2} < 9.491 \times 10^{-12}$ for every $C \in [17, 19]$. (Theorem 7.5.)

(ii) **Rank-independent off-diagonal bound.** The off-diagonal part of $A(C)$ defines a bounded operator on $\ell^2(\mathbb{Z})$ with norm below 19.836078. Every finite compression obeys the same bound. (Theorem 4.3, Proposition 4.2.)

(iii) **High-mode coercivity.** For every nonzero finitely supported x with $x_j = 0$ for $|j| < n_0$: $\langle x, A(C)x\rangle > \|x\|^2$ if x is even and $n_0 \ge 18{,}612{,}929$; and $\langle x, A(C)x\rangle > \|x\|^2$ for arbitrary x if $n_0 \ge 60{,}879{,}750$. (Theorem 6.4.)

The infinite range of indices in (i) and the rank independence in (ii) and (iii) are proved analytically (Sections 5–6). The only computer-assisted inputs are enclosures of finitely many real numbers: the boundary sum, sixteen signed moments and two absolute moments of $u(C)$, uniformly over the cell. Section 7 gives their validation theorems in full.

The decisive point is *order of operations*. The leading coefficient of R_n is $\sqrt{2}\,b_n S/n$, where $S = \sum_j c_j$ is the boundary value of the ground state. For the ground state $|S| < 6.6 \times 10^{-12}$ (Table 2). If S is formed before absolute values are taken, this cancellation survives. Expanding the remaining terms in signed moments preserves further cancellation. If instead each entry is bounded in absolute value and summed, the bound is 2.906, about 3×10^{11} times larger (Remark 5.6).

1.3 What is not proved

We state the limits of these results plainly, because the surrounding programme aims at RH.

- The theorems control the forcing at indices $j > 64$ and the far block $j \ge 18{,}612{,}929$. They do *not* control the inverse response of the intermediate modes $9 \le j < 18{,}612{,}929$. Eliminating those modes can amplify the forcing through a nearly singular constrained inverse: the certified ground eigenvalue lies in $[2.26 \times 10^{-26}, 1.38 \times 10^{-24}]$, and the spectral gap is about 4×10^{-23} (Section 8).
- No estimate here is uniform in growing support $C \to \infty$, and none provides a summable support/rank path.
- No limit is identified with Riemann's Ξ function, and the Riemann Hypothesis is not proved.

Figure 1 places these statements along the Fourier index.

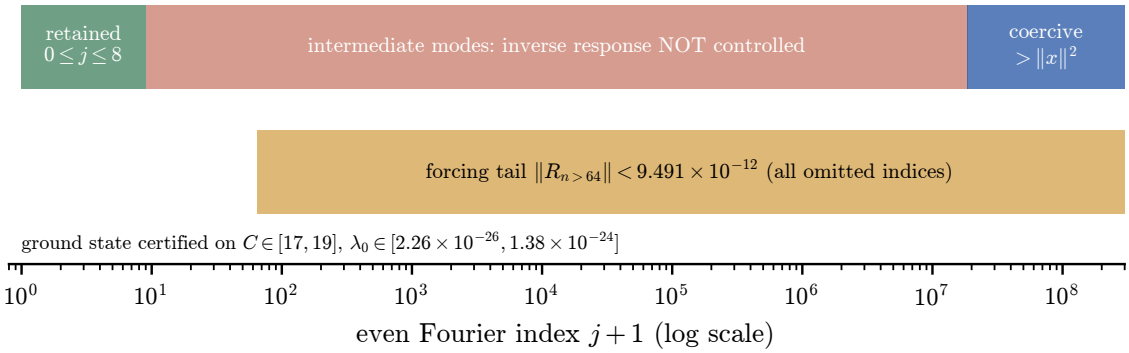


Figure 1: What is controlled along the even Fourier index, uniformly for $C \in [17, 19]$. Green: the retained rank-eight block, whose ground state is certified. Amber: the omitted-mode forcing, bounded for every index $n > 64$ by Theorem 7.5. Blue: the far block, on which the form exceeds the identity (Theorem 6.4). Rose: the intermediate modes, whose inverse response is not controlled here; this is the gap between the present results and any convergence statement.

1.4 Organization and conventions

Section 2 derives the matrix from the explicit-formula distributions. Section 3 evaluates its entries in closed form. Section 4 proves the uniform displacement bounds and the Hilbert-commutator estimate. Section 5 proves the omitted-mode identity and its infinite-index moment bound. Section 6 proves high-mode coercivity. Section 7 proves the validation lemmas and states the computer-assisted theorem. Section 8 records a consequence and the remaining obstructions. Appendix A reports an independent cross-check, and Appendix B gives replay instructions and hashes.

We use the following classical facts without proof: the Weierstrass product of Γ in the form (8), Gauss's integral for ψ, the expansion $E_1(x) = -\gamma - \log x + O(x)$ of the exponential integral [5, 5.7.6, 5.9.16, 6.6.2], Parseval's identity on the circle, and the inclusion property of interval arithmetic [7, 8]. Everything else is proved here.

2 The semilocal Weil form and its matrix

2.1 The explicit-formula distributions

For a function F on $\mathbb{R}_+^*$ we write $F^\vee(x) = F(x^{-1})$ and $d^*x = dx/x$. Following [9, 3], the distributions of the explicit formula in the normalization $F(x) = x^{1/2} f(x)$ are

$$W_{0,2}(F) = \int_0^\infty F(x)\,(x^{1/2} + x^{-1/2})\,d^*x, \tag{1}$$

$$W_p(F) = (\log p) \sum_{m \ge 1} p^{-m/2}(F(p^m) + F(p^{-m})), \tag{2}$$

$$W_{\mathbb{R}}(F) = (\log 4\pi + \gamma)F(1) + \int_1^\infty \Big(F(x) + F(x^{-1}) - 2x^{-1/2}F(1)\Big)\frac{x^{1/2}}{x - x^{-1}}\,d^*x. \tag{3}$$

Put $\Psi = W_{0,2} - W_{\mathbb{R}} - \sum_p W_p$. With the involutive convolution algebra $(f^* * g)(y) = \int \overline{f(x/y)}\,g(x)\,d^*x$ on $\mathbb{R}_+^*$, the Weil sesquilinear form is $QW(f, g) = \Psi(f^* * g)$.

Lemma 2.1 (Half-line reduction)**.** *Define on functions on* $[1,\infty)$

$$W_{0,2}^\sharp(F) = \int_1^\infty F(x)(x^{1/2}+x^{-1/2})\,d^*x, \qquad W_p^\sharp(F) = (\log p)\sum_{m\ge 1} p^{-m/2}F(p^m),$$

$$W_{\mathbb{R}}^\sharp(F) = \tfrac{1}{2}(\log 4\pi+\gamma)F(1) + \int_1^\infty \frac{x^{1/2}F(x)-F(1)}{x-x^{-1}}\,d^*x,$$

and $\Psi^\sharp = W_{0,2}^\sharp - W_{\mathbb{R}}^\sharp - \sum_p W_p^\sharp$*. Then* $\Psi(h) = \Psi^\sharp(h+h^\vee)$ *for every* h *for which both sides are defined.*

Proof. Each identity follows by splitting $\int_0^\infty$ at $x=1$ and substituting $x\mapsto x^{-1}$ on $(0,1)$. The measure d^*x is invariant under this substitution, and $x^{1/2}+x^{-1/2}$ is symmetric. For $W_{\mathbb{R}}$ note that $(h+h^\vee)(1) = 2h(1)$, which accounts for the factor $\frac{1}{2}$ in front of the constant and for the term $2x^{-1/2}F(1)$ in (3). □

2.2 Additive coordinates and the Fourier basis

Fix $C>1$ and $L=\log C$. The map $u\mapsto \log(C^{1/2}u)$ carries $[C^{-1/2},C^{1/2}]$ with d^*u isometrically onto $[0,L]$ with dx. It carries the multiplicative convolution to the additive one, $(f^**g)(y) = \int \overline{f(x-y)}\,g(x)\,dx$, and inversion to $y\mapsto -y$. We work on $\mathcal{H}_L = L^2([0,L],dx)$ with the orthonormal basis

$$U_n(x) = L^{-1/2}e^{iw_nx}, \qquad w_n = \frac{2\pi n}{L}, \qquad n\in\mathbb{Z}. \tag{4}$$

Set $q(f,g)(y) = (f^**g)(y) + (f^**g)(-y)$, an even function supported in $[-L,L]$. Pulled back to $[0,L]$ (where $y=\log x$, $x\ge 1$), the half-line distributions become

$$\mathfrak{D}(F) = \int_0^L F(y)\,2\cosh\tfrac{y}{2}\,dy - \mathfrak{A}(F) - \sum_{1<p^a<C}\frac{\log p}{p^{a/2}}F(a\log p), \tag{5}$$

$$\mathfrak{A}(F) = \tfrac{1}{2}(\log 4\pi+\gamma)F(0) + \int_0^\infty \frac{e^{y/2}F(y)-F(0)}{2\sinh y}\,dy, \tag{6}$$

for F continuous on $[0,\infty)$, piecewise C^1, and vanishing on $[L,\infty)$. A prime power with $p^a=C$ would contribute $F(L)=0$, so the sum may equally run over $p^a\le C$. In particular, on $17\le C\le 19$ the active set is $\{p^a\le 17\}$ throughout; at $C=19$ the term $p^a=19$ vanishes identically.

Lemma 2.2 (Correlations)**.** *For* $y\in[0,L]$*,*

$$q(U_m,U_n)(y) = \begin{cases} \dfrac{\sin(w_my)-\sin(w_ny)}{\pi(n-m)}, & m\ne n,\\[2mm] 2(1-\frac{y}{L})\cos(w_ny), & m=n.\end{cases}$$

Proof. For $0\le y\le L$, $(U_m^**U_n)(y) = L^{-1}\int_y^L e^{iw_m(y-x)}e^{iw_nx}\,dx$. If $m\ne n$ this equals $\big(e^{iw_my}-e^{iw_ny}\big)/(2\pi i(n-m))$, because $e^{i(w_n-w_m)L}=1$. If $m=n$ it equals $(1-y/L)e^{iw_ny}$. Since $(f^**g)(-y) = \overline{(g^**f)(y)}$ and the expression for $m\ne n$ is symmetric in (m,n), the function q is twice the real part. This gives the stated formulas. □

Definition 2.3 (Weil matrix)**.** The *Weil matrix* at support C is the infinite real matrix $A_{mn} = QW(U_m,U_n) = \mathfrak{D}\big(q(U_m,U_n)\big)$, for $m,n\in\mathbb{Z}$. Write $A = P - H - S$ for the pole, archimedean and prime contributions, i.e. the three terms of (5). For a finitely supported $c\in\mathbb{C}^{\mathbb{Z}}$ with $f=\sum_n c_nU_n$, one has $QW(f,f) = \langle c, Ac\rangle$.

Proposition 2.4 (Displacement structure)**.** *Let* $\mathrm{s}_n(y) = \sin(w_n y)\mathbf{1}_{[0,L]}(y)$ *and* $b_n = -\pi^{-1}\mathfrak{D}(\mathrm{s}_n)$*. Then* $b_{-n} = -b_n$*,* $b_0 = 0$*,* $A_{-m,-n} = A_{mn} = A_{nm}$*, and*

$$A_{mn} = \frac{b_m - b_n}{m - n} \quad (m \neq n), \qquad b_n = n\, A_{n0}. \tag{7}$$

The same holds separately for each of the three terms of (5)*, so that* $b = b^P + b^{\mathbb{R}} + b^{\Pi}$ *with each part odd in* n*.*

Proof. $\mathfrak{D}$ is linear, and $\mathrm{s}_n(0) = 0$, so (6) needs no regularization on s_n. By Lemma 2.2, $A_{mn} = \mathfrak{D}(\mathrm{s}_m - \mathrm{s}_n)/(\pi(n-m)) = (b_m - b_n)/(m-n)$. Oddness of $n \mapsto \mathrm{s}_n$ gives $b_{-n} = -b_n$ and $b_0 = 0$. Taking $n = 0$ gives $b_m = mA_{m0}$. The diagonal formula of Lemma 2.2 is even in n, and (7) is invariant under $(m, n) \mapsto (-m, -n)$. All three distributions are real on real functions. □

Remark 2.5. Equivalently, with $\mathcal{J} = \mathrm{diag}(n)$ and $e = (1)_n$, one has the commutator identity $[A, \mathcal{J}] = e\, b^T - b\, e^T$ on finitely supported vectors. This is the structure used in [3, Lemma 5.2].

Let γ be the reflection $U_n \mapsto U_{-n}$. Under the centring $x \mapsto x - L/2$ it is the reflection $x \mapsto -x$, which corresponds to $u \mapsto u^{-1}$ on $[C^{-1/2}, C^{1/2}]$. Since A commutes with γ, A preserves the even vectors $c_{-n} = c_n$. The even orthonormal basis is $e_0 = U_0$ and $e_j = (U_j + U_{-j})/\sqrt{2}$ for $j \geq 1$.

3 Closed formulas for the entries

3.1 Digamma facts

Let $\psi = \Gamma'/\Gamma$. We use

$$\psi(z) = -\gamma + \sum_{k\geq 0}\Big(\frac{1}{k+1} - \frac{1}{k+z}\Big), \qquad \psi'(z) = \sum_{k\geq 0}\frac{1}{(k+z)^2}, \tag{8}$$

$$\psi(z) = \int_0^\infty \Big(\frac{e^{-t}}{t} - \frac{e^{-zt}}{1-e^{-t}}\Big)dt, \qquad \psi'(z) = \int_0^\infty \frac{t\, e^{-zt}}{1-e^{-t}}\, dt \qquad (\mathrm{Re}\, z > 0). \tag{9}$$

The second identity in (9) follows from (8) by writing $(k+z)^{-2} = \int_0^\infty te^{-(k+z)t}dt$ and summing the geometric series, which Tonelli's theorem permits. From the first identity in (9) and $E_1(x) = \int_x^\infty e^{-t}t^{-1}dt = -\gamma - \log x + O(x)$,

$$\int_\varepsilon^\infty \frac{e^{-zt}}{1-e^{-t}}\, dt = -\gamma - \log\varepsilon - \psi(z) + o(1) \qquad (\varepsilon \downarrow 0). \tag{10}$$

Let $a_k = 2k + \frac{1}{2}$ and $\rho(y) = e^{y/2}/(2\sinh y)$. Then

$$\rho(y) = \sum_{k\geq 0} e^{-a_k y} \quad (y > 0), \qquad \int_\varepsilon^\infty \rho(y)e^{iwy}dy = \tfrac{1}{2}\int_{2\varepsilon}^\infty \frac{e^{-zt}}{1-e^{-t}}dt, \quad z = \tfrac{1}{4} - \tfrac{iw}{2}, \tag{11}$$

by the substitution $t = 2y$.

3.2 The three contributions

Write $D = \cosh(L/2) - 1$ and $z_n = \frac{1}{4} + i\pi n/L = \frac{1}{4} + iw_n/2$.

Proposition 3.1 (Pole). *For $m \neq n$, $P_{mn} = \dfrac{2D}{\pi(m-n)}\Big[\dfrac{w_m}{\frac{1}{4}+w_m^2} - \dfrac{w_n}{\frac{1}{4}+w_n^2}\Big]$, and $P_{nn} = \dfrac{4D}{L}\cdot\dfrac{\frac{1}{4}-w_n^2}{(\frac{1}{4}+w_n^2)^2}$.*

Proof. For $s = a + iw_n$ with $a = \pm\frac{1}{2}$ we have $e^{sL} = e^{aL}$, hence $\int_0^L e^{sy}\sin(w_n y)\,dy$ and the corresponding integral with $(1 - y/L)$ are elementary: $\int_0^L e^{ay}\sin(w_n y)\,dy = w_n(1-e^{aL})/(a^2+w_n^2)$ and $\int_0^L(1-\frac{y}{L})e^{sy}dy = -\frac{1}{s} + \frac{e^{aL}-1}{Ls^2}$. Summing over $a = \pm\frac{1}{2}$ with $2\cosh\frac{y}{2} = e^{y/2} + e^{-y/2}$ and taking real parts (the terms $-1/s$ cancel in pairs) gives $\int_0^L \mathrm{s}_n\, 2\cosh\frac{y}{2}\,dy = -2Dw_n/(\frac{1}{4}+w_n^2)$ and the stated diagonal. The off-diagonal formula follows from Proposition 2.4. □

Proposition 3.2 (Archimedean place). *For every $n \in \mathbb{Z}$,*

$$H_{nn} = \log\pi - \operatorname{Re}\psi(z_n) - \frac{\operatorname{Re}\psi'(z_n)}{2L} + \frac{2}{L}\sum_{k\geq 0} e^{-a_kL}\frac{a_k^2 - w_n^2}{(a_k^2+w_n^2)^2},$$

and $\mathfrak{A}(\mathrm{s}_n) = \frac{1}{2}\operatorname{Im}\psi(z_n) - \sum_{k\geq 0} e^{-a_kL}\dfrac{w_n}{a_k^2+w_n^2}$.

Proof. Let $F = 2(1-y/L)\cos(w_n y)\mathbf{1}_{[0,L]}$, so $F(0) = 2$. The integrand of (6) is bounded near 0 and integrable at ∞, so

$$\mathfrak{A}(F) = \log 4\pi + \gamma + \lim_{\varepsilon\downarrow 0}\Big[\int_\varepsilon^L \rho F\,dy - \int_\varepsilon^\infty \frac{dy}{\sinh y}\Big].$$

First, $\int_\varepsilon^\infty dy/\sinh y = \log\coth(\varepsilon/2) = -\log(\varepsilon/2) + O(\varepsilon^2)$. Second, split $\int_\varepsilon^L \rho F$ as $2\operatorname{Re}\int_\varepsilon^\infty \rho e^{iw_n y} - \frac{2}{L}\operatorname{Re}\int_\varepsilon^\infty y\rho e^{iw_n y} + 2\int_L^\infty \rho(\frac{y}{L}-1)\cos(w_n y)$. By (11) and (10) the first piece is $-\gamma - \log(2\varepsilon) - \operatorname{Re}\psi(\bar z_n) + o(1)$. By (9) with $t = 2y$ the second piece tends to $-\operatorname{Re}\psi'(\bar z_n)/(2L)$. In the third piece put $y = L + s$ and use $\cos(w_n(L+s)) = \cos(w_n s)$ together with (11): it equals $\frac{2}{L}\sum_k e^{-a_kL}\operatorname{Re}(a_k - iw_n)^{-2} = \frac{2}{L}\sum_k e^{-a_kL}(a_k^2-w_n^2)/(a_k^2+w_n^2)^2$. The constants combine to $\log 4\pi + \gamma + \log(\varepsilon/2) - \gamma - \log(2\varepsilon) = \log\pi$. Finally $\operatorname{Re}\psi(\bar z) = \operatorname{Re}\psi(z)$ and $\operatorname{Re}\psi'(\bar z) = \operatorname{Re}\psi'(z)$, and $H_{nn} = \mathfrak{A}(F)$.

For s_n we have $F(0) = 0$, so $\mathfrak{A}(\mathrm{s}_n) = \int_0^L \rho\,\mathrm{s}_n$. The imaginary part of (10) has a limit, and it gives $\operatorname{Im}\int_0^\infty \rho e^{iw_n y}dy = -\frac{1}{2}\operatorname{Im}\psi(\bar z_n) = \frac{1}{2}\operatorname{Im}\psi(z_n)$. The tail $\int_L^\infty \rho\sin(w_n y)\,dy$ equals $\sum_k e^{-a_kL}w_n/(a_k^2+w_n^2)$, by the same substitution. □

Proposition 3.3 (Primes). $S_{nn} = \sum_{p^a<C}\frac{\log p}{p^{a/2}}\,2(1-\frac{a\log p}{L})\cos(w_n a\log p)$, *and the prime contribution to b_n is $b_n^{\Pi} = \pi^{-1}\sum_{p^a<C}\frac{\log p}{p^{a/2}}\sin(w_n a\log p)$.*

Proof. Evaluate the finite sum in (5) on Lemma 2.2. □

Corollary 3.4 (Closed displacement sequence). *For $n \geq 1$,*

$$b_n = \frac{2D}{\pi}\frac{w_n}{\frac{1}{4}+w_n^2} + \underbrace{\frac{\operatorname{Im}\psi(z_n)}{2\pi} - \frac{1}{\pi}\sum_{k\geq 0} e^{-a_kL}\frac{w_n}{a_k^2+w_n^2}}_{=:\,b_n^{\mathbb{R}}} + \frac{1}{\pi}\sum_{p^a<C}\frac{\log p}{p^{a/2}}\sin(w_n a\log p). \tag{12}$$

Proof. Use $b_n = -\pi^{-1}\mathfrak{D}(\mathrm{s}_n)$ together with Propositions 3.1–3.3. The archimedean distribution enters $\mathfrak{D}$ with a minus sign, so it contributes $+\pi^{-1}\mathfrak{A}(\mathrm{s}_n)$. □

Remark 3.5. These closed forms are equivalent to the hypergeometric expressions of [3, Prop. 4.2–4.3]. The constant $\log\pi$ is obtained from the regularization, not fitted. Appendix A compares the closed entries with direct numerical quadrature of (5)–(6), using code that is independent of the certificate generator. They agree to better than 2×10^{-30}.

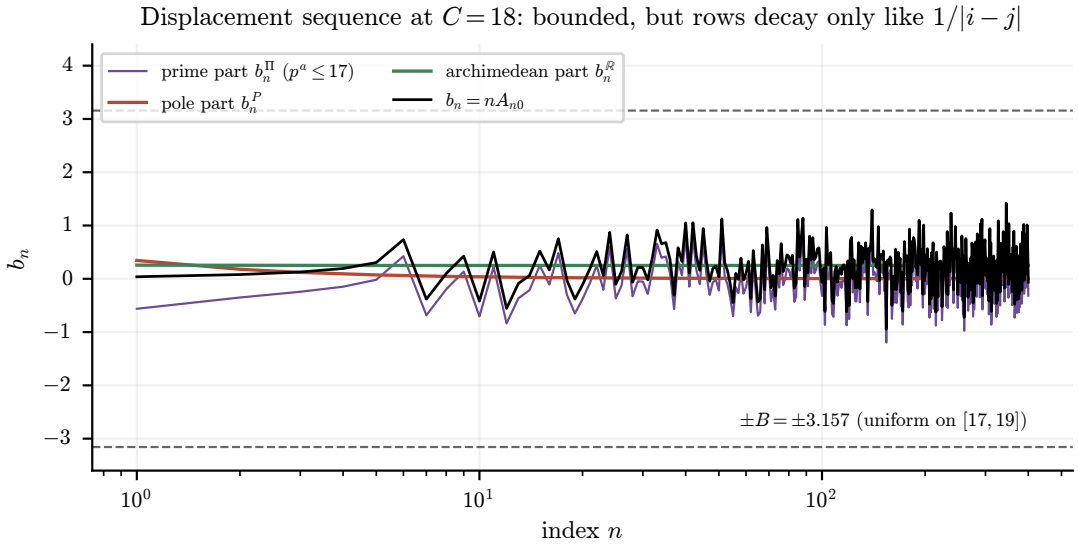


Figure 2: The displacement sequence $b_n = nA_{n0}$ at $C = 18$ and its three parts from Corollary 3.4, computed from the closed formulas. The pole part decays like $1/n$, the archimedean part tends to $1/4$, and the prime part oscillates without decay. The sequence stays inside the uniform bound $\pm B$ of Proposition 4.2, but $A_{mn} = (b_m - b_n)/(m-n)$ then decays only like $1/|m-n|$ along a row, which is why Section 4 needs Hilbert's inequality rather than row sums.

4 Rank-independent operator bounds

From now on $17 \le C \le 19$. Put $L_- = \log 17$, $L_+ = \log 19$, and

$$E = \frac{e^{-L_-/2}}{1-e^{-2L_-}}, \qquad \mathcal{P} = \sum_{p^a\le 17}\frac{\log p}{\sqrt{p^a}}, \qquad D_+ = \cosh\tfrac{L_+}{2} - 1. \tag{13}$$

Since $e^{L_-/2} = \sqrt{17}$ and $e^{L_+/2} = \sqrt{19}$, these are explicit algebraic and logarithmic numbers. Table 1 gives outward enclosures.

Lemma 4.1 (Digamma inequalities)**.** *For $y > 0$:*

(a) $0 < \operatorname{Im}\psi(\frac14 + iy) \le \frac{\pi}{2} + \min(2, 1/y)$;

(b) $\operatorname{Re}\psi(\frac14 + iy) \ge \log y - 1/y$;

(c) $|\psi'(\frac14 + iy)| \le 1/y^2 + \pi/(2y)$.

Proof. (a) By (8), $\operatorname{Im}\psi(\frac14+iy) = \sum_{k\ge0} g(k)$ with $g(t) = y/((t+\frac14)^2+y^2) > 0$ decreasing on $[0,\infty)$. Hence the sum is at most $g(0) + \int_0^\infty g = g(0) + \frac{\pi}{2} - \arctan\frac{1}{4y}$. Finally $g(0) = y/(\frac{1}{16}+y^2) \le \min(2, 1/y)$, since the maximum 2 is attained at $y = \frac14$.

(b) By (8) and $\gamma = \lim_M(\sum_{k<M}\frac{1}{k+1} - \log M)$, we have $\operatorname{Re}\psi(\frac14+iy) = \lim_M[\log M - \sum_{k<M} f(k)]$ with $f(t) = (t+\frac14)/((t+\frac14)^2+y^2)$. For each k, $|f(k) - \int_k^{k+1} f| \le \int_k^{k+1}|f'|$. The function f increases up to $t+\frac14 = y$ and then decreases, with maximum $1/(2y)$. Hence $\int_0^\infty |f'| \le 2\cdot\frac{1}{2y} = \frac1y$, and so $\sum_{k<M} f(k) \le \int_0^M f + \frac1y$. Since $\int_0^M f = \frac12\log\frac{(M+1/4)^2+y^2}{1/16+y^2}$, letting $M\to\infty$ gives $\operatorname{Re}\psi \ge \frac12\log(\frac{1}{16}+y^2) - \frac1y \ge \log y - \frac1y$.

(c) By (8), $|\psi'| \le \sum_k((k+\frac14)^2+y^2)^{-1}$. The $k=0$ term is at most y^{-2}. The sum of the rest is at most $\int_0^\infty dt/((t+\frac14)^2+y^2) \le \pi/(2y)$. □

Proposition 4.2 (Uniform displacement bounds)**.** *For all $C\in[17,19]$ and all $n\in\mathbb{Z}$,*

$$|b_n| \le B := \frac{2D_+}{\pi} + \frac{\frac{\pi}{2} + \min(2, L_+/\pi)}{2\pi} + \frac{E}{\pi} + \frac{\mathcal{P}}{\pi}, \tag{14}$$

$$|b_n^{\mathbb{R}}| \le B_{\mathbb{R}} := \frac{\frac{\pi}{2} + \min(2, L_+/\pi)}{2\pi} + \frac{E}{\pi}, \tag{15}$$

$$|b_n| \le B_0 + \frac{B_1}{|n|}\ (n\neq 0), \qquad B_0 = \frac14 + \frac{\mathcal{P}}{\pi}, \quad B_1 = \frac{L_+}{\pi^2}\Big(D_+ + \frac{1+E}{2}\Big). \tag{16}$$

Proof. By oddness it suffices to take $n\ge 1$; let $y = \pi n/L \ge \pi/L_+$. In (12): the pole term is at most $2D_+/\pi$, because $w/(\frac14+w^2)\le 1$, and also at most $2D_+/(\pi w_n) = D_+L/(\pi^2 n)$. The digamma term is controlled by Lemma 4.1(a), using $1/y \le L_+/\pi$ for (14) and $1/y = L/(\pi n)$ for (16). For the exponential series, $w/(a^2+w^2) \le 1/(2a) \le 1$ and also $\le 1/w$; moreover $\sum_k e^{-a_kL} = e^{-L/2}/(1-e^{-2L}) \le E$, because this function decreases in L. The prime sum has absolute value at most $\mathcal{P}/\pi$, because the active set is $\{p^a \le 17\}$ on the whole cell. Adding the pieces gives the three bounds. □

Theorem 4.3 (Hilbert commutator bound)**.** *Let $\beta\in\ell^\infty(\mathbb{Z})$ and let $T_{ij} = (\beta_i-\beta_j)/(i-j)$ for $i\neq j$, with $T_{ii}=0$. Then T extends from finitely supported vectors to a bounded operator on $\ell^2(\mathbb{Z})$ with $\|T\| \le 2\pi\|\beta\|_\infty$. For every set $I\subset\mathbb{Z}$, the compression $\Pi_I T\Pi_I$ obeys the same bound.*

Proof. Let $\mathsf{H}_{ij} = 1/(i-j)$ for $i\neq j$ and $\mathsf{H}_{ii}=0$. On $(0,2\pi)$ the function $\pi-\theta$ has Fourier coefficients $\frac{1}{2\pi}\int_0^{2\pi}(\pi-\theta)e^{-ik\theta}d\theta = -i/k$ for $k\neq 0$, and 0 for $k=0$, by one integration by parts. Hence $h(\theta) := \sum_{k\neq0} e^{ik\theta}/k = i(\pi-\theta)$ in $L^2(0,2\pi)$. For finitely supported x, y put $X(\theta) = \sum x_j e^{ij\theta}$ and $Y(\theta) = \sum y_j e^{ij\theta}$. Since $\overline{X}Y$ is a trigonometric polynomial,

$$\langle x, \mathsf{H}y\rangle = \sum_{i\neq j}\frac{\overline{x_i}y_j}{i-j} = \frac{1}{2\pi}\int_0^{2\pi} h\,\overline{X}Y\,d\theta,$$

so $|\langle x,\mathsf{H}y\rangle| \le \pi\cdot\frac{1}{2\pi}\int|X||Y| \le \pi\|x\|\|y\|$ by Cauchy–Schwarz and Parseval. Thus $\|\mathsf{H}\|\le\pi$; this is Hilbert's inequality [6, Ch. IX]. Now $T = D_\beta\mathsf{H} - \mathsf{H}D_\beta$ with $D_\beta = \operatorname{diag}(\beta_i)$, so $\|T\| \le 2\|\beta\|_\infty\|\mathsf{H}\|$. The compression is $D_{\Pi\beta}(\Pi\mathsf{H}\Pi) - (\Pi\mathsf{H}\Pi)D_{\Pi\beta}$, and $\|\Pi\mathsf{H}\Pi\|\le\pi$. □

Corollary 4.4. *On the whole cell, the off-diagonal part of $A(C)$ has norm at most $2\pi B < 19.836078$. The off-diagonal part of the archimedean contribution has norm at most $2\pi B_{\mathbb{R}} < 2.994796$.*

Individual rows of A_{off} are not absolutely summable, because $\sum_j |b_i-b_j|/|i-j|$ diverges logarithmically in general. Theorem 4.3 retains the signed structure that an absolute-row (Schur) bound discards.

Table 1: Constants for the cell $17 \le C \le 19$. Each enclosure is outward; the certificate (1024-bit Arb) and the independent check (mpmath interval arithmetic, 40 digits) agree.

Quantity	Definition	Enclosure
$\mathcal{P}$	(13)	$5.8323263624721337\ldots$
E	(13)	$0.2433777626232646\ldots$
B	(14)	< 3.157010
$B_{64} = B_0 + B_1/64$	(16)	< 2.115418
$2\pi B$	Cor. 4.4	< 19.836078
$2\pi B_{\mathbb{R}}$	Cor. 4.4	< 2.994796
$2\sinh(L_+/2) - L_+$	Lemma 6.1	< 1.185045

5 The omitted-mode identity and its infinite moment bound

Fix a retained rank $N \ge 1$ and real even coordinates $u = (u_0, \ldots, u_N)$. Its full coefficients are $c_0 = u_0$ and $c_{\pm i} = u_i/\sqrt{2}$; it has boundary sum $S = \sum_j c_j = u_0 + \sqrt{2}\sum_{i=1}^N u_i$. For $n > N$, Ac is even, so its component along e_n is $R_n := \langle e_n, Ac\rangle = \sqrt{2}(Ac)_n$, and its odd components vanish. Hence $\|\Pi_{|j|>J} Ac\|_{\ell^2(\mathbb{Z})} = \big(\sum_{n>J} R_n^2\big)^{1/2}$.

Lemma 5.1 (Paired identity)**.** *For every $n > N$,*

$$R_n = \frac{\sqrt{2}\, b_n S}{n} + \frac{2b_n}{n}\sum_{i=1}^{N} \frac{u_i\, i^2}{n^2 - i^2} - 2\sum_{i=1}^{N} \frac{i\, b_i u_i}{n^2 - i^2}. \tag{17}$$

Proof. By (7), $(Ac)_n = b_n \sum_j c_j/(n-j) - \sum_j b_j c_j/(n-j)$. Pair $j = \pm i$: $\frac{1}{n-i} + \frac{1}{n+i} = \frac{2n}{n^2-i^2}$; and, since $b_{-i} = -b_i$, $\frac{b_i}{n-i} - \frac{b_i}{n+i} = \frac{2ib_i}{n^2-i^2}$. The term $j = 0$ contributes u_0/n to the first sum and nothing to the second, because $b_0 = 0$. Finally write $\frac{2n}{n^2-i^2} = \frac{2}{n} + \frac{2i^2}{n(n^2-i^2)}$ and collect $\frac{\sqrt{2}}{n}(u_0 + \sqrt{2}\sum u_i) = \frac{\sqrt{2}S}{n}$. □

Definition 5.2 (Signed and absolute moments)**.** For an order $R \ge 0$, set $m_r = \sum_{i=1}^N u_i\, i^{2r}$ for $1 \le r \le R$, and $\nu_r = \sum_{i=1}^N u_i b_i\, i^{2r+1}$ for $0 \le r \le R-1$. Further set $M = \sum_{i=1}^N |u_i|\, i^{2R+2}$ and $V = \sum_{i=1}^N |u_i b_i|\, i^{2R+1}$.

Lemma 5.3 (Finite geometric expansion)**.** *For $n > N$,*

$$R_n = \frac{\sqrt{2} b_n S}{n} + 2b_n \sum_{r=1}^{R} \frac{m_r}{n^{2r+1}} - 2\sum_{r=0}^{R-1} \frac{\nu_r}{n^{2r+2}} + \mathcal{E}_n,$$

$$\mathcal{E}_n = \frac{2b_n}{n^{2R+1}} \sum_{i=1}^{N} \frac{u_i\, i^{2R+2}}{n^2 - i^2} - \frac{2}{n^{2R}} \sum_{i=1}^{N} \frac{u_i b_i\, i^{2R+1}}{n^2 - i^2}.$$

If $J > N$ and $|b_n| \le B_J$ for all $n > J$, then for all $n > J$

$$|\mathcal{E}_n| \le 2\gamma_J \Big[\frac{B_J M}{n^{2R+3}} + \frac{V}{n^{2R+2}}\Big], \qquad \gamma_J = \frac{1}{1 - (N/J)^2}.$$

Proof. With $x = i^2/n^2 < 1$ one has $\frac{x}{1-x} = \sum_{r=1}^R x^r + \frac{x^{R+1}}{1-x}$ and $\frac{1}{1-x} = \sum_{r=0}^{R-1} x^r + \frac{x^R}{1-x}$. Hence $\frac{i^2}{n^2-i^2} = \sum_{r=1}^R \frac{i^{2r}}{n^{2r}} + \frac{i^{2R+2}}{n^{2R}(n^2-i^2)}$ and $\frac{1}{n^2-i^2} = \sum_{r=0}^{R-1} \frac{i^{2r}}{n^{2r+2}} + \frac{i^{2R}}{n^{2R}(n^2-i^2)}$. Substitute these into (17). For the bound, note that $i \le N < J < n$ gives $\frac{1}{n^2-i^2} = \frac{n^{-2}}{1-(i/n)^2} \le \frac{\gamma_J}{n^2}$. □

Lemma 5.4 (Integral test)**.** *For integers $p \geq 1$ and $J \geq 1$, $H_p(J) := \left(\sum_{n>J} n^{-2p}\right)^{1/2} \leq J^{1/2-p}/\sqrt{2p-1}$.*

Proof. $t \mapsto t^{-2p}$ decreases, so $\sum_{n>J} n^{-2p} \leq \int_J^\infty t^{-2p} dt$. □

Theorem 5.5 (Infinite-index tail bound)**.** *Let $N \geq 1$, $R \geq 0$, $J > N$, and $|b_n| \leq B_J$ for $n > J$. For every real even $u \in \mathbb{R}^{N+1}$,*

$$\Big(\sum_{n>J} R_n^2\Big)^{1/2} \leq \sqrt{2} B_J |S| H_1 + 2B_J \sum_{r=1}^{R} |m_r| H_{2r+1} + 2\sum_{r=0}^{R-1} |\nu_r| H_{2r+2} + 2\gamma_J \big[B_J M H_{2R+3} + V H_{2R+2}\big], \quad (18)$$

with $H_p = H_p(J)$.

Proof. Apply Minkowski's inequality in $\ell^2(\{n > J\})$ to the decomposition of Lemma 5.3. Bound each sequence pointwise by a constant times n^{-p}; then Lemma 5.4 gives its ℓ^2 norm. □

Theorem 5.5 is a statement about every omitted index at once. It holds for any finite rank, any cutoff $J > N$ and any order R. Neither the sum over n nor the remainder is truncated. The only inputs it needs are the scalars S, m_r, ν_r, M, V and B_J.

Remark 5.6 (Why the signs matter)**.** Bounding $|A_{nj}| \leq 2B/|n-j|$ entrywise and summing gives the *uncoupled* estimate $\left(\sum_{n>J} R_n^2\right)^{1/2} \leq 2\sqrt{2}B(|u_0| + \sqrt{2}\sum|u_i|)/\sqrt{J-N}$. For the rank-eight ground on the cell this is about 2.906 when $J = 64$. Theorem 5.5 gives 9.49×10^{-12} (Theorem 7.5), roughly 3.06×10^{11} times smaller. The improvement comes from three facts: the ground state has a tiny boundary sum S, low signed moments m_1 and ν_0 that are small, and the moment expansion preserves the sign cancellation across the retained coordinates.

6 High-mode coercivity

Lemma 6.1 (Pole form)**.** *For $f \in \mathcal{H}_L$, let $\tilde{f}(x) = f(x + L/2)$ on $[-L/2, L/2]$. Then $\langle c, Pc\rangle = 2|\langle \cosh \frac{\cdot}{2}, \tilde{f}\rangle|^2 - 2|\langle \sinh \frac{\cdot}{2}, \tilde{f}\rangle|^2$. Consequently $P \geq 0$ on even vectors, and $P \geq -(2\sinh \frac{L}{2} - L)$ on all vectors.*

Proof. By (1), in additive coordinates, $W_{0,2}(f^* * f) = \iint \overline{f(s)} f(t)\, 2\cosh \frac{s-t}{2}\, ds\, dt$. This kernel is translation invariant, so we may centre. Then $2\cosh \frac{s-t}{2} = 2\cosh \frac{s}{2} \cosh \frac{t}{2} - 2\sinh \frac{s}{2} \sinh \frac{t}{2}$. The two functions are orthogonal on $[-L/2, L/2]$. For even f the sinh term vanishes. In general the negative eigenvalue is $-2\|\sinh \frac{\cdot}{2}\|^2 = -(2\sinh \frac{L}{2} - L)$. □

Lemma 6.2 (Prime form)**.** *$|\langle c, Sc\rangle| \leq 2\mathcal{P}\|c\|^2$ on the cell.*

Proof. $\langle c, Sc\rangle = \sum_{p^a < C} \frac{\log p}{p^{a/2}} (F(a \log p) + F(-a \log p))$ with $F(y) = \int \overline{f(x-y)} f(x)\, dx$, and Cauchy–Schwarz gives $|F(y)| \leq \|f\|^2$. □

Lemma 6.3 (Archimedean diagonal)**.** *For $n \neq 0$ and $C \in [17, 19]$,*

$$-H_{nn} \geq \delta(|n|) := \log \frac{|n|}{L_+} - \frac{L_+}{\pi|n|} - \frac{1}{4|n|} - \frac{L_+(1+E)}{2\pi^2 n^2}.$$

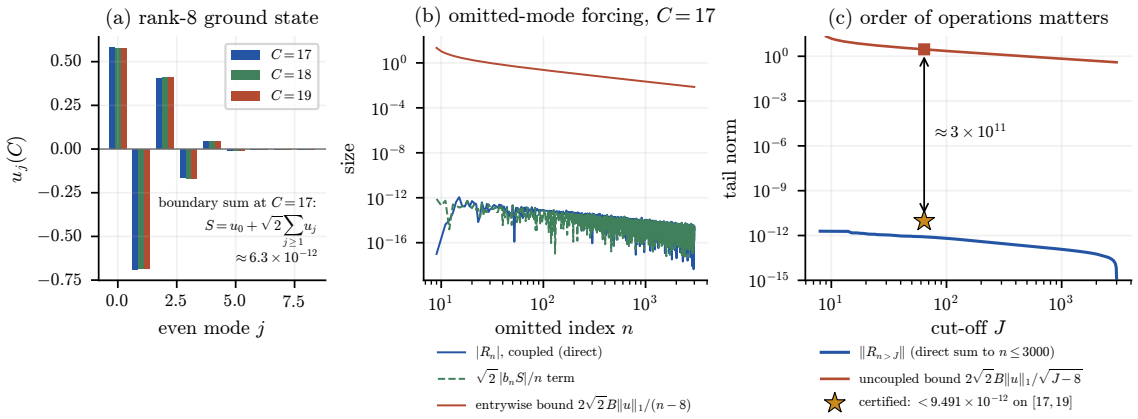


Figure 3: The cancellation behind Theorem 7.5, recomputed in 40-digit floating point (illustration, not certificate). (a) The unit even ground vector $u(C)$ at three supports; its boundary sum S is of order 10^{-12}. (b) The exact omitted-mode forcing $|R_n|$ at $C = 17$ by direct summation of the retained terms, its leading S-term, and the entrywise bound obtained by taking absolute values first. (c) The tail norm $\|R_{n>J}\|$ summed directly to $n \leq 3000$, the uncoupled bound of Remark 5.6, and the certified bound at $J = 64$. The arrow is the factor $\approx 3 \times 10^{11}$ gained by forming S and the signed moments before taking absolute values.

Proof. Use Proposition 3.2 with $y = \pi|n|/L$. By Lemma 4.1(b), $-\log\pi + \operatorname{Re}\psi(z_n) \geq \log\frac{|n|}{L} - \frac{L}{\pi|n|}$. By Lemma 4.1(c), $\frac{|\psi'(z_n)|}{2L} \leq \frac{L}{2\pi^2 n^2} + \frac{1}{4|n|}$. The exponential series is at most $\frac{2}{L}\sum_k e^{-a_k L} w_n^{-2} \leq \frac{LE}{2\pi^2 n^2}$, because $|a^2 - w^2| \leq a^2 + w^2$. Each resulting term is monotone in $L \in [L_-, L_+]$ in the direction used. □

Theorem 6.4 (High-mode coercivity)**.** *For $n_0 \geq 1$ put*

$$\beta_{\mathrm{even}}(n_0) = \delta(n_0) - 2\pi B_{\mathbb{R}} - 2\mathcal{P}, \qquad \beta_{\mathrm{full}}(n_0) = \beta_{\mathrm{even}}(n_0) - (2\sinh\tfrac{L_+}{2} - L_+).$$

For every $C \in [17, 19]$ and every finitely supported x with $x_j = 0$ for $|j| < n_0$, one has $\langle x, A(C)x\rangle \geq \beta_{\mathrm{full}}(n_0)\|x\|^2$. If moreover x is even, then $\langle x, A(C)x\rangle \geq \beta_{\mathrm{even}}(n_0)\|x\|^2$. Both lower bounds increase to $+\infty$ as n_0 grows, and

$$\beta_{\mathrm{even}}(18{,}612{,}929) > 1.000000028, \qquad \beta_{\mathrm{full}}(60{,}879{,}750) > 1.000000002.$$

Proof. Decompose $\langle x, A\rangle x = \langle x, P\rangle x - \langle x, H_{\mathrm{diag}}x\rangle - \langle x, H_{\mathrm{off}}x\rangle - \langle x, S\rangle x$. Then apply Lemma 6.1, Lemma 6.3 (with δ increasing in $|n|$), Corollary 4.4 (compressions keep the bound) and Lemma 6.2. The numerical values come from outward interval evaluation (Table 1). The certificate uses 1024-bit Arb; the independent check uses mpmath intervals. Both report $\beta_{\mathrm{even}}(18{,}612{,}928) < 1$ and $\beta_{\mathrm{full}}(60{,}879{,}749) < 1$. Thus the stated starts are the least integers for which *these particular* bounds exceed one; they are not claimed sharp for A itself. □

Since the span of the U_n is a form core for the closed semilocal form [3, Prop. 3.4], the same lower bounds hold for the closure of the far-block form. We use only finite compressions below.

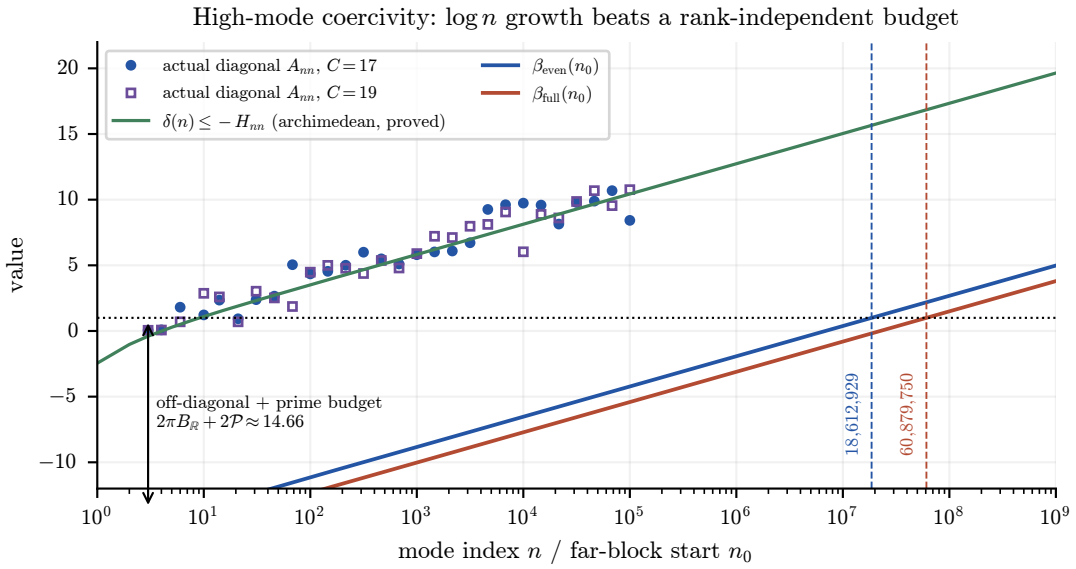


Figure 4: Mechanism of Theorem 6.4. Markers: the actual diagonal entries A_{nn} at $C = 17$ and $C = 19$, from the closed formulas. Green: the proved archimedean lower bound $\delta(n)$ of Lemma 6.3, which grows like $\log n$. Blue and red: the coercivity constants $\beta_{\mathrm{even}}(n_0)$ and $\beta_{\mathrm{full}}(n_0)$, which subtract the rank-independent off-diagonal and prime budget (and, for β_{full}, the negative pole eigenvalue). They cross 1 exactly at the thresholds stated in the theorem; the thresholds reflect these particular bounds, not the true onset of coercivity of A.

7 The finite-rank certificate

Theorem 5.5 reduces the infinite tail to finitely many real numbers attached to $u(C)$. This section proves that the enclosures of those numbers are valid uniformly in $C \in [17, 19]$. All arithmetic is midpoint–radius ball arithmetic with outward rounding (Arb via python-flint 0.8.0, 1024 bits [7]). The *inclusion property* is the only fact about the computer we use: each computed ball contains the exact value of the corresponding real expression for every choice of inputs from the input balls.

7.1 Analytic Taylor enclosure in the support parameter

Replace $\operatorname{Re}\psi(z_n)$ by $\frac{1}{2}[\psi(\frac{1}{4} - \frac{i\pi n}{L}) + \psi(\frac{1}{4} + \frac{i\pi n}{L})]$, and $\operatorname{Im}\psi$ by the corresponding odd combination divided by $2i$. Then every entry of Section 3 is holomorphic in L on $\operatorname{Re} L > 0$, for a fixed active prime set. Let $c_* = \frac{1}{2}(L_- + L_+)$, $\delta = \frac{1}{2}(L_+ - L_-) < 0.055613$ and $R_* = \frac{1}{4}$.

Lemma 7.1 (Cauchy remainder)**.** *Let W be holomorphic on $|L - c_*| \le R_*$ with $|W| \le M_W$ there, and let W_j be its Taylor coefficients at c_*. If $|h| \le \delta < R_*$ and $q = \delta/R_*$, then $|W(c_* + h) - \sum_{j \le d} W_j h^j| \le M_W q^{d+1}/(1 - q)$.*

Proof. Cauchy's estimate gives $|W_j| \le M_W/R_*^j$; sum the geometric tail. □

The certificate [1, artifact `weil_support_taylor_17_19_8_degree64`] encloses every coefficient of the 9×9 even block to degree $d = 64$. It truncates the exponential series after 80 terms, with an omitted tail bounded uniformly over the complex disk and over *all* index pairs. It bounds M_W on the circle by covering it with rectangles. The resulting operator-norm error of the polynomial matrix over the real cell is below 4.870×10^{-41}.

7.2 Validated eigenfamily

The cell is covered by 32 adjacent closed subcells. On each subcell the polynomial matrix is translated to the local centre. A candidate eigenbasis $U(L)$ and candidate eigenvalues $\Lambda = \operatorname{diag}(\ell_i(L))$ are polynomials with exact dyadic coefficients, written in a fixed orthogonal centre basis Q. No convergence of the recurrence that proposed them is assumed.

Lemma 7.2 (Uniform isolation)**.** *Let $B = Q^T A Q$ be real symmetric. Let $E = BU - U\Lambda$, and suppose $\|U - I\|_\infty \le \beta < 1$. Put $\eta = \|E\|_\infty/(1-\beta)$. If the intervals $[\ell_i - \eta, \ell_i + \eta]$ are pairwise disjoint, then each contains exactly one eigenvalue of A, in increasing order.*

Proof. By the Neumann series, $\|U^{-1}\|_\infty \le 1/(1-\beta)$, and $U^{-1}BU = \Lambda + U^{-1}E$. Gershgorin's theorem with row sums places the spectrum in the union of the disks of radius η about the ℓ_i. By continuity of eigenvalues along $t \mapsto \Lambda + tU^{-1}E$, a union of k disjoint disks contains exactly k eigenvalues. The eigenvalues are real because A is symmetric. □

Lemma 7.3 (Eigenvector distance)**.** *Let A be real symmetric with simple lowest eigenvalue λ_0 and unit eigenvector v. Let $\hat{u}$ be a unit vector with $\|A\hat{u} - \ell\hat{u}\| \le \rho$, and suppose every other eigenvalue satisfies $\lambda - \ell \ge D > 0$. If v is oriented so that $\langle v, \hat{u}\rangle \ge 0$, then $\|\hat{u} - v\| \le \sqrt{2}\rho/D$.*

Proof. Write $\hat{u} = \alpha v + w$ with $w \perp v$ and $\alpha \ge 0$. Projecting $(A - \ell)\hat{u}$ onto $v^\perp$ gives $(A - \ell)w$, so $D\|w\| \le \rho$. Then $\|\hat{u} - v\|^2 = 2(1-\alpha) \le 2(1-\alpha^2) = 2\|w\|^2$. □

All 32 subcells pass Lemma 7.2, with $\|U - I\|_\infty < 0.01633$ and disk radii below 4.456×10^{-40}. An independent odd-block certificate with the same subdivision places every odd eigenvalue above the even ground. Hence $u(C)$ is the simple ground of the full 17×17 truncation; the tail theorem below does not need this. Uniformly on the cell, the lowest eigenvalue lies in $[2.259 \times 10^{-26}, 1.380 \times 10^{-24}]$, and the full ground gap exceeds 4.043×10^{-23}. Lemma 7.3 gives

$$\|\hat{u}(C) - u(C)\| \le d < 8.687 \times 10^{-21}, \tag{19}$$

where $\hat{u} = p/\|p\|$ is the normalized candidate. The candidate's centre-basis anchor coordinate is identically one, so $\|p\| \in [1, \bar{\nu}]$ for a certified $\bar{\nu}$. The unit ground origin satisfies $u_0 > 0.5753$, which fixes the orientation consistently across subcells.

7.3 Enclosures of the moments

Lemma 7.4 (Linear functionals of the exact ground)**.** *Let $\omega \in \mathbb{R}^{N+1}$. For C in a subcell,*

$$\langle \omega, u(C)\rangle \in \frac{\operatorname{range}\langle \omega, p\rangle}{[1, \bar{\nu}]} + [-d\|\omega\|,\, d\|\omega\|].$$

If the weights depend on C through $b_i(L) = \operatorname{poly}_i(L) + \eta_i$ with $|\eta_i| \le \epsilon_i$, then

$$\sum_i \omega_i b_i u_i \in \frac{\operatorname{range}\sum_i \omega_i \operatorname{poly}_i p_i}{[1, \bar{\nu}]} + \Big[\pm\Big(d\,\big\|(\omega_i \sup|b_i|)_i\big\| + \sum_i |\omega_i|\epsilon_i \sup|p_i|\Big)\Big].$$

Proof. Write $u = \hat{u} + (u - \hat{u})$, use (19) with Cauchy–Schwarz, and use $\|p\| \ge 1$ for the polynomial-error term. □

The key design choice is that each moment is formed as *one* polynomial in the support parameter, namely $\sum_i \omega_i p_i(L)$ or $\sum_i \omega_i b_i(L) p_i(L)$, *before* its range is enclosed. Neighbouring Fourier coordinates nearly cancel, and a common parameter is shared, so enclosing each coordinate separately would destroy the cancellation that Remark 5.6 quantifies. We apply the lemma with:

- $\omega = (1, \sqrt{2}, \dots, \sqrt{2})$ for S. The spatial certificate evaluates S as the endpoint Bernstein coefficient of the spatial ground kernel, which equals S exactly;
- $\omega_i = i^{2r}$ $(i \geq 1)$ for m_r, $1 \leq r \leq 8$;
- $\omega_i = i^{2r+1}$ with $b_i(L) = iA_{i0}(L)$ taken from the same Taylor certificate, for ν_r, $0 \leq r \leq 7$;
- absolute weights after the signed expansion, for M and V with $R = 8$.

7.4 The main computer-assisted theorem

Theorem 7.5 (Coupled infinite Fourier tail)**.** *For every $C \in [17, 19]$, the unit even rank-eight ground vector $u(C)$ satisfies*

$$\Big(\sum_{n>64} R_n(C)^2\Big)^{1/2} \leq 9.4900466 \times 10^{-12} < 9.491 \times 10^{-12}.$$

Proof. On each of the 32 subcells, apply Theorem 5.5 with $N = 8$, $J = 64$, $R = 8$ and $B_{64} = B_0 + B_1/64$; this is valid because $n > 64$ implies $|b_n| \leq B_0 + B_1/n < B_{64}$ by Proposition 4.2. The scalar inputs are enclosed uniformly over the subcell by Lemma 7.4. The right side of (18) is evaluated in outward ball arithmetic, and the bound is the maximum over the subcells. Table 2 lists the contributions for the worst subcell, which is adjacent to $C = 17$. □

Table 2: Worst subcell (index 0, adjacent to $C = 17$) of Theorem 7.5. Each entry is a certified outward upper bound. The moment intervals are certified enclosures over the whole subcell.

Term of (18)	Certified input	Contribution
$\sqrt{2}B_{64}\lvert S\rvert H_1$	$S \in [6.3167,\ 6.5920] \times 10^{-12}$	2.465×10^{-12}
$2B_{64}\lvert m_1\rvert H_3$	$m_1 \in [-6.317, -5.976] \times 10^{-8}$	3.648×10^{-12}
$2B_{64}\lvert m_2\rvert H_5$	$m_2 \in [1.1354,\ 1.1920] \times 10^{-4}$	1.252×10^{-12}
$2B_{64}\lvert m_3\rvert H_7$	$m_3 \in [-7.691, -7.419] \times 10^{-2}$	1.642×10^{-13}
$2B_{64}\sum_{r=4}^{8}\lvert m_r\rvert H_{2r+1}$		$< 9.5 \times 10^{-15}$
$2\lvert\nu_0\rvert H_2$	$\nu_0 \in [-5.406, -4.750] \times 10^{-10}$	1.219×10^{-12}
$2\lvert\nu_1\rvert H_4$	$\nu_1 \in [1.5495,\ 1.7150] \times 10^{-6}$	6.182×10^{-13}
$2\lvert\nu_2\rvert H_6$	$\nu_2 \in [-1.5124, -1.3961] \times 10^{-3}$	1.062×10^{-13}
$2\sum_{r=3}^{7}\lvert\nu_r\rvert H_{2r+2}$		$< 7.9 \times 10^{-15}$
remainder $2\gamma_{64}[\cdots]$	M, V	1.001×10^{-22}
total		9.4900×10^{-12}

Over all 32 subcells the bound ranges from 3.03×10^{-12} to 9.490×10^{-12}, and the expansion remainder never exceeds 1.219×10^{-22}. The uncoupled estimate of Remark 5.6 ranges from 2.889 to 2.906.

8 A consequence, and what remains open

Corollary 8.1 (Far-block inverse energy)**.** *Let $n_0 = 18{,}612{,}929$. Let Q_M be the orthogonal projection onto* $\operatorname{span}\{e_j : n_0 \le j \le M\}$ *for any $M \ge n_0$, let $\lambda(C)$ be the lowest even eigenvalue at rank eight, and let $T_M = Q_M A Q_M - \lambda I$ on the range of Q_M. Then, uniformly in M and $C \in [17, 19]$,*

$$\langle Q_M Ac, T_M^{-1} Q_M Ac\rangle \le \frac{\|Q_M Ac\|^2}{1-\lambda} < 9.007 \times 10^{-23}.$$

Proof. By Theorem 6.4, $Q_M A Q_M \ge \beta_{\mathrm{even}}(n_0) > 1$, so $T_M \ge 1 - \lambda$, with $0 < \lambda < 1.380 \times 10^{-24}$. Since $n_0 > 64$, Theorem 7.5 gives $\|Q_M Ac\|^2 < (9.49005 \times 10^{-12})^2 < 9.0061 \times 10^{-23}$. □

This is the energy of the *direct* far-block response. It is not the change of the ground state when the rank is increased. The following problems remain open.

Open Problem 8.2 (Intermediate block)**.** Control the Schur complement obtained by eliminating the modes $9 \le j < 18{,}612{,}929$, together with the profile response it induces. The forcing into those modes is not bounded here; for $9 \le j \le 64$ it is a finite computation, but for $j > 64$ only its ℓ^2 norm is controlled. More importantly, the eliminated block contains spectrum close to the ground eigenvalue. Tiny energy does not imply a tiny profile increment; the response can be amplified by a nearly singular inverse. A Davis–Kahan bound [4] that uses the full residual would need a spectral gap of the infinite operator larger than the forcing, whereas the certified finite gap is about 4×10^{-23}.

Open Problem 8.3 (All supports and ranks)**.** Find estimates of Theorem 7.5 type that are uniform as $C \to \infty$ and $N \to \infty$ jointly, with a summable combined support/rank path. The present constants $\mathcal{P}$, D_+ and the coercivity start n_0 all grow with C, so the argument does not extend without new ideas.

Open Problem 8.4 (Identification with Ξ)**.** Show that suitably normalized ground profiles converge to Riemann's Ξ function, as proposed in [3, §7]. Nothing in this paper addresses that identification.

Consequently, the results here are finite-cell, rank-independent estimates. They are not, and do not imply, a proof of the Riemann Hypothesis.

Theorem-status ledger

Statement	Status
Lemmas 2.1–2.2, Props. 2.4–3.3, Cor. 3.4	Proved (from the explicit-formula distributions)
Lemma 4.1, Prop. 4.2, Thm. 4.3	Proved; constants by interval evaluation
Lemmas 5.1–5.4, Thm. 5.5	Proved, all ranks, all indices
Thm. 6.4	Proved; thresholds by interval evaluation
Lemmas 7.1–7.4	Proved
Thm. 7.5, Cor. 8.1	Proved, computer-assisted (validated ball arithmetic)
Appendix A	Independent floating-point evidence, not proof
Problems 8.2–8.4, RH	Open

Software and use of AI tools

Ball arithmetic uses Arb through python-flint 0.8.0 [7]; the independent cross-check uses mpmath. The author used AI tools to write and execute the code for the certificate generators, the regression tests and the independent cross-check. Every computational claim in this paper is reproducible from the replay commands in Appendix B.

A Independent cross-check

The script `paper/weil_tail_crosscheck.py` shares no code with the certificate generator. It implements the formulas of Sections 3–5 directly in mpmath at 80 digits. Its results, which are floating-point evidence and not part of any proof:

1. *Closed entries versus the defining distributions.* At $C = 17$ the closed entries $(0,0)$, $(3,3)$, $(8,8)$, $(1,0)$, $(5,-2)$, $(8,-8)$ and $(7,6)$ agree with adaptive quadrature of (5)–(6) to within 1.6×10^{-30}. This checks the $\log\pi$ regularization of Proposition 3.2 independently.
2. *Paired identity.* At $C = 17, 18, 19$ and $n = 9, 65, 1000$, (17) agrees with a direct product $\sqrt{2}\sum_j A_{nj}c_j$ to within 10^{-60}.
3. *Ground moments.* The independently computed unit ground at $C = 17$, 18 and 19 gives

C	S	m_1	ν_0	m_2	ν_1
17	6.3425×10^{-12}	-6.2547×10^{-8}	-5.3873×10^{-10}	1.1860×10^{-4}	1.7116×10^{-6}
18	1.3773×10^{-12}	-1.9552×10^{-8}	-1.2350×10^{-10}	5.2081×10^{-5}	5.3834×10^{-7}
19	4.0274×10^{-12}	-4.1936×10^{-8}	-3.4992×10^{-10}	8.3493×10^{-5}	1.1650×10^{-6}

   In each case S, m_1 and ν_0 lie inside the certified intervals of the containing subcell (indices 0, 16 and 31).
4. *Direct omitted-mode sum.* Summing R_n^2 directly for $64 < n \le 20{,}000$ and bounding $n > 20{,}000$ by Theorem 5.5 with $R = 4$ gives totals $\le 1.02 \times 10^{-12}$, 2.02×10^{-13} and 6.45×10^{-13} at $C = 17, 18, 19$. These are consistent with the certified uniform bound, and the direct sum at $C = 17$ (8.21×10^{-13}) shows that the bound is conservative by a factor of about 11.
5. *Constants.* The mpmath interval evaluation reproduces every entry of Table 1. It also reproduces both coercivity thresholds, including the failure at $n_0 - 1$.

B Replay and data

The certificate chain (matrix Taylor, even eigenfamily, odd block, spatial boundary, rank tail) is stored in `research/prime_spectral/`. Each artifact records the SHA-256 hashes of its source files and of its input artifacts, and the generators refuse stale inputs. The tail certificate `weil_rank_tail_cell_17_19_8_degree64.json` records the input hashes

Taylor	`5f745aa6175bc5b06477d90e587f2435b74feae4a1012af70710bacaa573e0db`
even	`8a977827a832ccaca8c706eab1ad334e4636d2508518b21f3e708a6a39a1083b`

Replay the tail certificate (7 tests, including a full replay of the 32-cell cover) and the independent check with

```
uv run --with python-flint==0.8.0 --python 3.12 python -m unittest \
  discover -s experiments/prime_spectral -p test_weil_rank_tail_cell.py -v
uv run --with mpmath python paper/weil_tail_crosscheck.py 20000
```

The full prime-spectral regression suite (159 tests) is run with the same command and the pattern `test_*.py`. The analytic arguments of this paper are not formalized in Lean.

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